p-value method + Excel execution

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

- Asked for p-value or Excel decision for a mean test.
- This sheet focuses on the in-scope two-tail Z route.
- p-value measures extremeness assuming H_0.

2 DECISION / METHOD

1. Compute zcalc under H_0.
2. For two tails, double the upper-tail area beyond |zcalc|.
3. Compare p-value with α.
4. p<α -> reject H_0; otherwise do not reject H_0.
5. State contextual evidence.

3 FORMULA

```
two-tail p = 2*[1-Φ(|zcalc|)]
reject H0 if p-value < α
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: Z lane and SE.
Still know: p-value definition, two-tail doubling, decision comparison and conclusion language.

5 EXCEL ROUTE

- `n=COUNT(range); Xbar=AVERAGE(range)`
- `zcalc=(Xbar-mu0)/(sigma/SQRT(n))`
- `=(1-NORM.S.DIST(ABS(zcalc),TRUE))*2.`

6 TINY WORKED EXAMPLE

If zcalc=-2.10, two-tail p is about .0357. At α=.05, reject H_0.

7 TOP EXAM TRAPS

- p-value=P(H_0 true).
- Forgetting ABS or the factor 2.
- Rejecting when p>α.
- Reporting 'accept H_0'.

8 SAY ALOUD

The p-value is how extreme the sample is under H_0; below α means reject H_0.

9 EXIT CHECK

Write the exact Excel formula for a two-tail Z p-value from zcalc in B11.

ONE-GLANCE VISUAL ROUTE

